

Complex Number Operations

Answer Key

Algebra 2

Quick Answers

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|---------------|-------------------------|
| 1. $10 + 12i$ | 7. the product = 29, — |
| 2. $5 - 10i$ | 8. $-7 + 24i$ |
| 3. $2 - 8i$ | 9. $-7 + 32i$ |
| 4. $15 + 12i$ | 10. $5 - 21i$ |
| 5. $11 + 10i$ | 11. -10 |
| 6. $13 - 9i$ | 12. (a) $= 1 + 21i$, — |
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What is verified

12 of 14 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 7, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSN-CN.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

7. If they got 21: treated i squared as $+1$ instead of -1

1. Simplify $(8 + 3i) + (2 + 9i)$.

Solution: Adding complex numbers means adding the real parts and the imaginary parts separately, because i behaves like a variable that is being collected: $(8 + 2) + (3 + 9)i$.

$10 + 12i$

2. Simplify $(7 - 4i) - (2 + 6i)$.

Solution: Distribute the subtraction over both parts of the second number first: $7 - 4i - 2 - 6i$. Now collect: $(7 - 2) + (-4 - 6)i$.

$5 - 10i$

3. Simplify $(-3 + 5i) + (9 - 11i) - (4 + 2i)$.

Solution: Distribute the subtraction over the last pair, then collect real with real and imaginary with imaginary.

$$\begin{aligned} &(-3 + 5i) + (9 - 11i) - (4 + 2i) \\ &= (-3 + 9 - 4) + (5 - 11 - 2)i \end{aligned}$$

The real parts give 2 and the imaginary parts give -8 .

$2 - 8i$

4. Multiply and write in standard form: $3i(4 - 5i)$.

Solution: Distribute the monomial, then replace i^2 by -1 .

$$\begin{aligned} 3i(4 - 5i) &= 12i - 15i^2 \\ &= 12i - 15(-1) \\ &= 15 + 12i \end{aligned}$$

Notice that multiplying by i moved the real given into the imaginary part and the imaginary given into the real part.

$15 + 12i$

5. Multiply and write in standard form: $(2 + 3i)(4 - i)$.

Solution: Multiply out all four products the way you would with two binomials, then use $i^2 = -1$.

$$\begin{aligned} (2 + 3i)(4 - i) &= 8 - 2i + 12i - 3i^2 \\ &= 8 + 10i - 3(-1) \\ &= 11 + 10i \end{aligned}$$

$11 + 10i$

6. Simplify $(10 - 2i) - (-3 + 7i)$.

Solution: Subtracting a negative real part adds: $10 - (-3) = 13$, and $-2 - 7 = -9$ for the imaginary part.

$13 - 9i$

7. Multiply $(5 - 2i)(5 + 2i)$, then describe what kind of number the product is and why the i terms cancelled.

Solution: Multiply out all four products.

$$\begin{aligned} (5 - 2i)(5 + 2i) &= 25 + 10i - 10i - 4i^2 \\ &= 25 - 4(-1) \\ &= 29 \end{aligned}$$

29

Written part (instructor-judged). A correct description says the product is a *real* number, and explains why: the two middle terms $+10i$ and $-10i$ are opposites, so they add to zero, and the remaining $-4i^2$ becomes $+4$. In general $(a - bi)(a + bi) = a^2 + b^2$, which is always real. Award no credit for repeating the number without saying why the i terms disappear.

8. Expand and simplify $(3 + 4i)^2$.

Solution: A square is still a product of two binomials — the middle term is what students most often lose.

$$\begin{aligned}(3 + 4i)^2 &= 9 + 12i + 12i + 16i^2 \\ &= 9 + 24i + 16(-1) \\ &= -7 + 24i\end{aligned}$$

$-7 + 24i$

9. Multiply and write in standard form: $(-6 + i)(2 - 5i)$.

Solution: The second factor of the first number is $1i$, so the products are $-6 \cdot 2$, $-6 \cdot (-5i)$, $i \cdot 2$ and $i \cdot (-5i)$.

$$\begin{aligned}(-6 + i)(2 - 5i) &= -12 + 30i + 2i - 5i^2 \\ &= -12 + 32i + 5 \\ &= -7 + 32i\end{aligned}$$

$-7 + 32i$

10. Simplify $2(4 - 3i) - 3(1 + 5i)$.

Solution: Distribute each real multiplier first; no i^2 appears because neither factor is imaginary.

$$\begin{aligned}2(4 - 3i) - 3(1 + 5i) &= 8 - 6i - 3 - 15i \\ &= 5 - 21i\end{aligned}$$

$5 - 21i$

11. Simplify $(2 - 3i)^2 + (2 + 3i)^2$.

Solution: Expand each square separately, keeping the middle terms.

$$\begin{aligned}(2 - 3i)^2 &= 4 - 12i + 9i^2 = -5 - 12i \\ (2 + 3i)^2 &= 4 + 12i + 9i^2 = -5 + 12i\end{aligned}$$

The imaginary parts are opposites, so they cancel when the two squares are added: $-5 + (-5) = -10$.

-10

12. Let $z = 3 - 2i$ and $w = -1 + 5i$. (a) Compute $zw - 2z$. (b) Explain why the product of two numbers that each carry an imaginary part can end up with a real part neither factor had.

Solution (a): Multiply first, then subtract $2z$.

$$zw = (3 - 2i)(-1 + 5i) = -3 + 15i + 2i - 10i^2 = 7 + 17i$$

$$2z = 6 - 4i$$

$$zw - 2z = (7 - 6) + (17 + 4)i$$

$1 + 21i$

Written part (b) — instructor-judged. A full-credit answer names two moves: the two imaginary terms multiply to give a term containing i^2 , and replacing i^2 with -1 turns that term into a real number, which then combines with the product of the two real parts. Here $(-2i)(5i) = -10i^2 = +10$, and $-3 + 10 = 7$ — a real part larger than either factor's. Half credit for naming only one of the two moves.